

Emergency Maintenance Report —

EMR-2025-001

APEX EMERGENCY GRID SOLUTIONS

24/7 High-Voltage Rapid Response · NAESB-Certified Emergency Contractor
1450 Industrial Parkway, Tacoma, WA 98421 · Dispatch: (253) 555-0177

Field	Value
Document ID	EMR-2025-001
Issued By	Apex Emergency Grid Solutions
Report Date	2025-05-09
Site	NW-PowerPool-WIND-014
Asset	WIN-10058 (GE wind turbine, 3.6 MW class)
Outage ID	OUT-2025-0433
Cascadia Work Order	WO-2025-05033
Cascadia Authorization	EPA-2024-11 (Emergency Procurement Authorization invoked 2025-05-02 14:18 PDT)
Apex Crew Lead	R. Halverson, Lead HV Technician (NABCEP #41827)
Distribution	Cascadia Operations · Cascadia Vendor Management · Cascadia AP

1. Scope of Emergency Work

Apex was dispatched 2025-05-02 at 14:18 PDT after Cascadia Power Services (CPS) reported a >6-hour mobilization estimate. The forced outage was declared on WIN-10058 at 11:52 PDT following a state-of-charge drift alarm cascading into inverter-side under-voltage trip. Scope of authorized emergency work:

- Onsite stabilization and lock-out/tag-out of WIN-10058 at NW-PowerPool-WIND-014.
- Diagnostic sweep of battery management module (BMM), inverter capacitor bank, and DC-link.

- Restoration to service or recommendation of follow-on capital work if restoration not feasible within EPA-2024-11 scope.

Outage duration to restoration: **227 hours**. Lost generation reported by Cascadia SCADA: 2,506.8 MWh.

2. Findings

On arrival (2025-05-02 22:40 PDT), Apex crew documented the following:

1. **Inverter capacitor bank C-2** showed measured ESR of 412 mΩ against an OEM allowable of 180 mΩ. Three of nine capacitors visibly bulged; dielectric oil weeping observed at C-2-04 and C-2-07.
2. **Battery management module BMM-A** logged 41 cell-balancing faults in the preceding 14 days, none of which had triggered SCADA alarms above advisory level. Firmware version 3.1.2, last updated 2023-08; current OEM-recommended baseline is 3.4.1.
3. **DC-link bus voltage** ripple measured at 6.8% peak-to-peak against OEM spec of <2.5%. This ripple is the proximate cause of the SoC drift signature observed by the operator.
4. The capacitor bank, BMM, and DC-link form an integrated subsystem on this GE design. In Apex's field judgment, the degradation pattern is **irreversible**: capacitor electrolyte loss has occurred over a multi-year window and the BMM firmware/hardware combination cannot be brought back into OEM tolerance through component-level swaps alone.

3. Repairs Performed

Emergency stabilization and partial restoration:

Item	Detail	Hours / Qty
Onsite diagnostics, fault tree walk	Lead + 2 technicians	38 hrs
Capacitor replacement, partial (C-2-04, C-2-07, C-2-09)	Apex PN AP-CAP-3300-DC	3 units
BMM firmware re-flash to 3.4.1	Includes 2 failed flash attempts	22 hrs
Re-energization sequence	3 attempted, 2 failed on residual harmonic	47 hrs
Re-coordination of GE inverter protection settings	Per GE Manual VTM-GE-R3.1	11 hrs

Item	Detail	Hours / Qty
Standby / fault wait time (weather + harmonic decay)	Weekend, 1.5× multiplier	41 hrs
Travel and onsite logistics	2 vehicles, 1 crane assist	—
Total labor		187 hours

Material: 3× Apex PN AP-CAP-3300-DC; 1× BMM firmware module (loaned/returned, no charge); 14× balance-of-system consumables.

4. Recommendations

The partial capacitor replacement performed under this emergency authorization will keep WIN-10058 in service for an estimated 90–180 days. However, in Apex's professional judgment:

1. The capacitor bank degradation is symptomatic of broader inverter and battery management module failure. We recommend **full inverter assembly and BMM replacement** within 120 days. Estimated capital scope: \$1.05M (separate proposal AP-Q-2025-118 to follow).
2. Continued operation on the partially restored bank will produce repeat SoC drift events. Apex cannot warrant restoration beyond the 90-day window.
3. The DC-link ripple condition is consistent with what GE Service Advisory GE-SA-2022-14 describes as "irreversible degradation" of this generation of inverter platform.
4. We note the firmware version (3.1.2) was overdue for upgrade by more than 18 months. Apex recommends Cascadia review preventive cadence on this platform. (Reference Cascadia internal CAR-2024-118 cited in the dispatch packet.)

5. Invoice Summary

Line	Rate	Qty	Extended
Labor — emergency rate, standard hours	\$420.00/hr	146 hrs	\$61,320.00
Labor — emergency rate, weekend 1.5×	\$630.00/hr	41 hrs	\$25,830.00
Subtotal labor			\$87,150.00
Parts — capacitors AP-CAP-3300-DC	\$14,200 ea	3	\$42,600.00

Line	Rate	Qty	Extended
Parts — consumables, dielectric, terminals	itemized attachment A	—	\$11,840.00
Emergency parts mark-up	35% on parts subtotal	—	\$19,054.00
Crane / lift services (subcontract)	pass-through + 12%	—	\$28,400.00
Travel, per-diem (3 technicians × 8 days)	per Apex tariff	—	\$14,200.00
Mobilization premium (EPA <4hr dispatch)	flat	1	\$42,000.00
Engineering review, OEM coordination	\$185/hr	26	\$4,810.00
EPA-2024-11 emergency response surcharge	30% on labor	—	\$26,145.00
Documentation, post-energization test report	flat	—	\$6,800.00
Subtotal			\$283,000.00
Risk / availability premium (per Apex MSA §4.3)	—	—	\$129,000.00
Total Invoice			\$412,000.00

Closing

Equipment restored to limited service 2025-05-12 at 06:43 PDT. Apex recommends Cascadia engineering schedule the proposed inverter/BMM replacement promptly. Photographs of failed capacitors (12 images), DC-link oscilloscope traces, and firmware flash logs are appended as Attachment A through D.

Apex stands ready to mobilize the recommended capital scope under separate quotation.

Signed	Name	Date
Crew Lead	R. Halverson	2025-05-09
Apex Engineering Review	D. Pieteron, PE	2025-05-09
Apex Account Manager	J. Beaumont	2025-05-09

Attachments: Photographs (12), DC-link oscilloscope traces (4), firmware flash logs, Apex proposal AP-Q-2025-118 (inverter/BMM replacement, separate cover).

Apex references: GE Service Advisory GE-SA-2022-14 · GE Manual VTM-GE-R3.1 · Apex SOP AEX-SOP-INV-04.

Cascadia references cited in dispatch packet: EPA-2024-11 · CAR-2024-118 · OIR (in preparation).